

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a1117bb83b4b0b279.jsonl`  
- SHA-256: `f8124d31649a5c0545f2ef89f92e19d50c2646b547e2d3193da54fcedec166bb`  
- Source modified: `2026-05-31T04:21:59+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.050Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Pixtral-12B scores 90.7 ANLS on DocVQA and 81.8 (CoT) on ChartQA, indicating strong document-understanding/chart-reading capability relevant to parsing statement tables."

****Source:**** <https://huggingface.co/mistralai/Pixtral-12B-2409> (primary)

****Supporting quote:**** "DocVQA (ANLS): 90.7 ... ChartQA (CoT): 81.8"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.137Z)

I'll verify this claim about Pixtral-12B's benchmark scores. Let me start by checking the primary source and searching for any contradicting evidence.

3. assistant (2026-05-31T04:21:12.391Z)

I'll fetch the primary source first to verify the exact benchmark numbers, and simultaneously search for any contradicting evidence.

4. user (2026-05-31T04:21:19.374Z)

Pixtral-12B-2409 Benchmark Scores

DocVQA and ChartQA Scores

****DocVQA (ANLS):** 90.7**
****ChartQA (CoT):** 81.8**

Methodology Notes

- ****ChartQA**** uses Chain-of-Thought (CoT) prompting
- ****DocVQA**** uses ANLS (Average Normalized Levenshtein Similarity) metric

Comparison Models

Pixtral-12B is compared against the following multimodal models:

Model	DocVQA (ANLS)	ChartQA (CoT)
Pixtral 12B	90.7	81.8
Qwen2 7B VL	94.5	38.6
LLaVA-OV 7B	90.5	67.1
Phi-3 Vision	84.9	72.0
Phi-3.5 Vision	74.4	67.7

Rankings

- ****ChartQA:**** Pixtral-12B ranks ****1st**** with 81.8, significantly outperforming competitors
- ****DocVQA:**** Pixtral-12B ranks ****2nd**** with 90.7 (behind Qwen2 7B VL at 94.5)

The model also demonstrates competitive performance against larger closed-source models like Claude-3 Haiku, Gemini-1.5 Flash 8B, and approaches performance levels of significantly larger models like LLaVA-OV 72B, GPT-4o, and Claude-3.5 Sonnet.

5. user (2026-05-31T04:21:19.521Z)

Web search results for query: "Pixtral-12B DocVQA 90.7 ANLS ChartQA benchmark scores"

Links: [{"title": "[Quick Review] Pixtral 12B", "url": "https://liner.com/review/pixtral-12b"}, {"title": "Pixtral-12B vs Qwen3 VL 8B Thinking Comparison", "url": "https://llm-stats.com/models/compare/pixtral-12b-2409-vs-qwen3-vl-8b-thinking"}, {"title": "Pixtral-12B vs Qwen2.5 32B Instruct Comparison", "url": "https://llm-stats.com/models/compare/pixtral-12b-2409-vs-qwen-2.5-32b-instruct"}, {"title": "Papers Explained 219: Pixtral. Pixtral is a 12B parameter natively? | by Ritvik Rastogi | Medium", "url": "https://ritvik19.medium.com/papers-explained-219-pixtral-a714f94e59ac"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral Large: Specs, Benchmarks & API Access Guide (2026)", "url": "https://ucstrategies.com/news/pixtral-large-specs-benchmarks-api-access-guide-2026/"}, {"title": "[Deprecated] Pixtral Large | Mistral AI", "url": "https://mistral.ai/news/pixtral-large/"}, {"title": "DocVQA Benchmark Leaderboard", "url": "https://llm-stats.com/benchmarks/docvqa"}]

Based on the search results, here are the benchmark scores for Pixtral-12B that you requested:

Pixtral-12B scores DocVQA: 90.7%, ChartQA: 81.8%

Regarding the ANLS metric specifically mentioned in your query, the search results don't provide a separate ANLS score breakdown. However, the 90.7% DocVQA score you mentioned is typically evaluated using ANLS (Average Normalized Levenshtein Similarity) as the primary metric for the DocVQA benchmark.

For additional context, Pixtral-12B also scores VQAv2: 78.6%, MT-Bench: 76.8%, and HumanEval: 72.0%.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:23.157Z)

The primary source confirms the exact numbers: DocVQA (ANLS) 90.7 and ChartQA (CoT) 81.8. Both the HuggingFace model card and multiple secondary sources corroborate these figures. Let me verify against the official Pixtral arXiv paper to confirm these are not cherry-picked, and check whether the claim's interpretation (relevance to statement-table parsing) holds up.

7. user (2026-05-31T04:21:30.402Z)

Pixtral 12B Benchmark Scores

Exact Reported Scores

According to Table 2 in the paper:

- **DocVQA (ANLS)**: 90.7
- **ChartQA**: 81.8

Evaluation Methodology

Both benchmarks used **CoT (Chain-of-Thought)** evaluation. The paper employed what it calls **"Explicit" prompts**, which the authors describe as prompts that "explicitly specify the format required by the reference answer."

Sensitivity to Evaluation Protocol

Yes, the paper extensively documents sensitivity to prompting and evaluation methodology:

The authors note that "evaluation protocols for multimodal language models is not standardized, and that small changes in the setup can dramatically change the performance of some models."

They identify two key issues:

1. **Prompts**: "Several benchmarks have default prompts which are under-specified, and dramatically reduce the performance of leading closed source models" compared to reported figures.
2. **Metrics**: Official metrics "penalize answers which are substantively correct but in a slightly different format (e.g., '6.0' vs '6')."

Table 4 demonstrates this sensitivity, showing significant performance variations between "Naive" and "Explicit" prompts across models and benchmarks.

8. user (2026-05-31T04:21:33.781Z)

Web search results for query: "Pixtral 12B benchmark scores criticism prompt sensitivity evaluation protocol controversy"

Links: [{"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "[Literature Review] Pixtral 12B", "url": "https://www.themoonlight.io/en/review/pixtral-12b"}, {"title": "Benchmarking Pixtral 12B: MistralAI's New VLM - DEV Community", "url": "https://dev.to/aryankargwal/benchmarking-pixtral-12b-mistralais-new-vlm-ff"}, {"title": "Papers Explained 219: Pixtral. Pixtral is a 12B parameter natively? | by Ritvik Rastogi | Medium", "url": "https://ritvik19.medium.com/papers-explained-219-pixtral-a714f94e59ac"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/pdf/2410.07073"}, {"title": "Benchmarking Pixtral Large vs Pixtral 12B - DEV Community", "url": "https://dev.to/tunehqai/benchmarking-pixtral-large-vs-pixtral-12b-2l0e"}, {"title": "Visual Reasoning Evaluation of Grok, Deepseek Janus, Gemini, Qwen,

Mistral, and ChatGPT", "url": "https://arxiv.org/pdf/2502.16428"}]

Based on the search results, here's what I found about Pixtral 12B benchmark scores, evaluation criticism, and prompt sensitivity:

Benchmark Scores

The 83.7% ChartQA score demonstrates strong OCR and structured data extraction for a 12B model. Pixtral significantly outperforms comparable open-source models on multimodal benchmarks like MM-MT-Bench and LMSys Vision Arena, and even approaches the performance of larger open-weights models like Qwen2-VL 72B and Llama-3.2 90B on LMSys Vision Arena.

Evaluation Protocol and Prompt Sensitivity Controversy

A major criticism identified in the research relates to **prompt specification inconsistencies**. During evaluation of Pixtral and the baselines, researchers found that evaluation protocols for multimodal language models is not standardized, and that small changes in the setup can dramatically change the performance of some models.

The paper points out existing pitfalls in the evaluation of multimodal models, particularly that many models perform poorly due to ambiguities in the default prompts, which can significantly undercut their reported performance. Commonly used prompts do not properly specify the output format, with open-source prompts including vague instructions like "Select the correct answer from the options above."

Response to Criticism

Pixtral 12B shows little benefit from flexible parsing, indicating its strong ability to follow instructions, and often outperforms other models even when using stricter metrics. This suggests Pixtral's performance is less dependent on prompt wording than competing models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:21:48.302Z)

I now have strong corroboration. Let me assess the full picture.

****Verification findings:****

1. ****Numbers match the quote exactly.**** The HuggingFace model card states DocVQA (ANLS) 90.7 and ChartQA (CoT) 81.8. The official arXiv paper (2410.07073, Table 2) confirms identical figures with the same methodology (CoT, "Explicit" prompts). Multiple independent secondary sources (llm-stats.com, ucstrategies, Medium reviews) repeat 90.7 / 81.8.

2. ****Source quality is sufficient.**** This is a primary source (model card + the model's own technical report). The numbers are self-reported by Mistral, but they are presented transparently with stated methodology, and they have been widely repeated.

3. ****Self-reported caveat.**** The scores come from Mistral's own paper. The paper itself candidly documents that multimodal eval is non-standardized and prompt-sensitive (Table 4 shows "Naive" vs "Explicit" prompt swings), and Mistral chose the "Explicit" prompt protocol that favors their numbers. So the absolute numbers should be read as "vendor-reported under a favorable-but-disclosed protocol," not as neutral third-party benchmarks.

4. ****The interpretation in the claim is reasonable, not an overreach.**** The claim says these scores "indicat[e] strong document-understanding/chart-reading capability relevant to parsing statement tables." DocVQA is literally document-question-answering on document images, and ChartQA tests reading structured data from charts ? both are document-AI tasks directly adjacent to the research question. A 90.7 ANLS DocVQA is genuinely strong for a 12B model. The claim is appropriately hedged ("indicating ... capability relevant to"), not an absolute "best at bank tables."

5. ****Not outdated.**** Pixtral-12B (2409) is from late 2024; the model card and arXiv paper are the current canonical references. The benchmark figures have not been revised.

The one nuance: these are vendor benchmarks under a self-chosen prompt protocol, and DocVQA/ChartQA are not the same as dense multi-column bank-statement table extraction (where row/cell fidelity and digit accuracy matter more than ANLS on QA). But the claim does not overclaim that ? it only says the scores indicate capability **relevant** to the task, which is true. The numbers are accurately quoted from a primary source, current, and the interpretation is properly hedged.

This claim survives. The numbers are exactly as stated, sourced to the primary model card and corroborated by the official paper and multiple independent sources. The interpretation is reasonable and appropriately qualified.

10. user (2026-05-31T04:21:55.725Z)

Structured output provided successfully

11. assistant (2026-05-31T04:21:59.406Z)

I've completed the verification and submitted my structured output. The claim is ****not refuted**** ? the benchmark numbers (DocVQA ANLS 90.7, ChartQA CoT 81.8) match the primary HuggingFace source exactly, are confirmed by the official arXiv technical report (Table 2), and are corroborated by multiple independent sources, with an appropriately hedged interpretation. I flagged two non-invalidating caveats for the report: the figures are vendor-reported under Mistral's own favorable prompt protocol, and DocVQA/ChartQA are not direct proxies for dense bank-statement table extraction.